

Beyond Surfactants: Janus Particles for Functional Interfaces and Coatings

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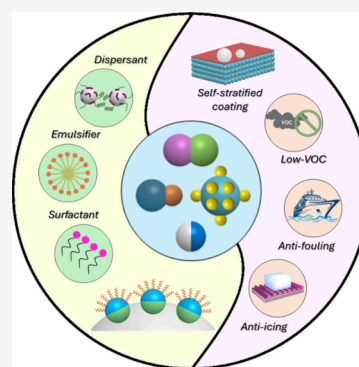
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ABSTRACT: Janus particles (JPs), initially introduced as soft matter, have evolved into a distinctive class of materials that set them apart from traditional surfactants, dispersants, and block copolymers. This mini-review examines the similarities and differences between JPs and their molecular counterparts to elucidate the unique properties of JPs. Key studies on the assembly behavior of JPs in bulk phases and at interfaces are reviewed, highlighting their unique ability to form diverse, complex structures. The superior interfacial stability and tunable amphiphilicity of JPs make them highly effective emulsifiers and dispersants, particularly in emulsion polymerization systems. Beyond these applications, JPs demonstrate immense potential as coating materials, facilitating the development of eco-friendly, anti-icing, and antifouling coatings. A comparative discussion with zwitterionic polymers also highlights the distinctive advantages of each system. This review emphasizes that while JPs mimic some of the behaviors of small molecular surfactants, they also open doors to entirely new applications, making them indispensable as next-generation functional materials.



1. INTRODUCTION

The concept of Janus particles (JPs) was introduced by P.G. de Gennes in his 1991 Nobel laureate lecture.¹ Named after the ancient Roman god, these particles exhibit distinct chemical properties on each side. In his lecture, de Gennes presented four examples of materials: polymers, surfactants, liquid crystals, and Janus grains. Although these materials may seem unrelated at first glance, they do share a unifying characteristic—anisotropy. This asymmetric feature enables them to assemble into diverse structures and transition through various phases, as their interactions depend not only on their positions but also on their orientations. Furthermore, de Gennes emphasized “complexity” and “flexibility” as defining qualities for these materials, coining the term “Soft Matter”.

Following de Gennes’ ideas, amphiphilic JPs, with one side hydrophobic and the other hydrophilic, closely resemble surfactant molecules. This similarity prompts the question of whether JPs exhibit behaviors similar to surfactants. De Gennes speculated that JPs might adsorb at liquid–liquid interfaces, forming a “skin that can breathe.” However, de Gennes’ ideas could not be experimentally validated until much later, as producing high-quality JPs with diverse chemistries and precise geometry was a nontrivial task. In the early years, research on JPs was limited due to synthetic challenges, with most initial efforts focused on theoretical calculations and simulations.²

Over time, various synthetic routes have been developed, starting with straightforward chemistries and small-scale approaches, such as directional coating, and advancing to more sophisticated approaches, including copolymer-based methods and seeded emulsion polymerization.^{3–6} As synthesis improved to produce JPs with consistent size and defined geometry, experimental studies increased, substantially expanding the understanding of their assembly structures and adsorption behaviors in bulk and at interfaces.⁷

In many cases, amphiphilic JPs indeed exhibit behaviors similar to small surfactant molecules. Early studies on micron-sized, half-coated amphiphilic JPs demonstrated that at low concentrations, these particles could form small clusters resembling micellar structures, with their hydrophobic sides oriented inward toward each other.^{8,9} At higher concentrations, they assemble into elongated, worm-like structures, similar to the rod-like micelles formed by surfactants with specific hydrophilic–lipophilic balance (HLB) values, a

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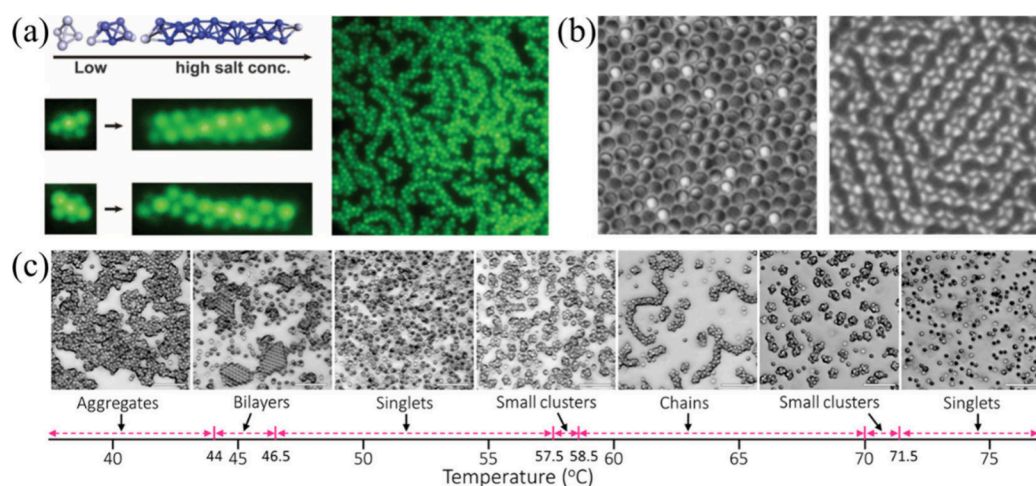


Figure 1. (a) Salt concentration-mediated self-assembly of JPs, transitioning from small clusters at low salt levels to elongated structures and interconnected networks at higher concentrations, demonstrating tunable supracolloidal architectures.¹⁰ (Reproduced with permission from ref 10. Copyright 2011, Science.) (b) Two-dimensional hexagonally packed crystalline structures formed by spherical amphiphilic JPs, exhibiting long-range stripe orientations as hydrophobic sides align.¹⁹ (Reproduced with permission from ref 19. Copyright 2014, American Physical Society.) (c) Temperature-dependent phase transitions and reconfigurable self-assembly of DNA-coated JPs, showing formation of aggregates, bilayers, singlets, small clusters, and chains as temperature varies from 40 to 75 °C.¹⁶ (Reproduced with permission from ref 16. Copyright 2020, American Chemical Society.)

measure of hydrophobicity.^{10–13} Recent studies on DNA-coated particles and smaller JPs have shown assemblies with even closer structural similarities to those formed by surfactant molecules, such as capsules, vesicles, and bilayers (Figure 1).^{14–17} However, significant differences exist between surfactant molecules and JPs, partly due to their substantial size disparity, which leads to contrasting interactions and diffusion rates. Studies on micron-sized JPs indicate that their kinetics is significantly influenced by their colloidal geometry (Figure 1).^{10,18} More intriguingly, spherical amphiphilic JPs can form two-dimensional hexagonally packed crystalline structures, with orientations that display long stripes overlaying the positional order as their hydrophobic sides orient toward each other (Figure 1c).¹⁹ This showcases the unique interplay between colloidal characteristics and amphiphilicity.²⁰ Recent studies further reveal that JPs interact distinctively with ionic liquids, adopting specific orientations near substrates.²¹

Due to the much larger size of JPs, their adsorption at interfaces differs significantly from that of small surfactant molecules. Calculations indicate that JPs with optimized geometries exhibit much higher adsorption energies than those of homogeneous particles, highlighting their potential as efficient emulsifiers.²² However, studies on interfacial adsorption using micron-sized JPs suggest that interactions among JPs play a significant role in their behavior. These interactions often arise from surface chemistry and capillary forces created by interface pinning at particle boundaries.^{23,24} Additionally, adsorption barriers such as particle charges may hinder JPs from approaching the interfaces.²⁵

Owing to their colloidal nature, JPs can readily be enhanced with various functionalities. For instance, they can be made magnetic, allowing manipulation by external magnetic fields, or coated with metal layers to produce unique electrophoretic flows and introduce more complex interactions among particles.^{15,26–29} Additionally, catalytic surface modifications can yield “active particles” driven by chemical reactions.^{30–32} Studies on these functionalities and their broad applications have been recently reviewed elsewhere.³³ Here we will focus

on applications utilizing the assembly of amphiphilic JPs, especially in improving coating materials, with analogies drawn to surfactants, block copolymers, and dispersants. We will also briefly compare JPs to zwitterionic polymers, as both incorporate opposing properties within a single entity.

This review starts by covering the basic physicochemical properties and assembly behaviors of JPs, followed by an exploration of their applications, particularly in roles similar to those of surfactants and dispersants. This approach highlights key parallels and deepens our understanding of JPs. Studies have shown that JPs can not only replicate the functions of their molecular counterparts but also offer significant advantages in stabilizing emulsions and creating more sustainable and functional coatings. Given the close relationship between synthesis and application, we will discuss these aspects together, with a focus on synthesis methods suited to each application. Although various synthetic techniques exist, none are universally applicable. We will also elaborate on emulsion polymerization, a commonly used method adopted in the coatings industry for producing sustainable, waterborne coatings.

2. ASSEMBLY STRUCTURES IN BULK

The assembly structures of surfactants are influenced by their HLB value, and a similar concept governs JPs through the *Janus balance*, which can be simply viewed as a measure of the amphiphilicity of JPs. A more rigorous definition of Janus balance involves the adsorption energy of JPs at the interface, accounting for their surface chemistry and geometry.³⁴ By determining the preferred assembly structures, Janus balance enables JPs to self-organize into various configurations, much like how the HLB value guides surfactant assembly. For instance, micron-sized JPs can form clusters and chain-like structures, or even complex bilayers and vesicles depending on their concentration and interactions.¹⁶ These principles remain relevant even as the size of JPs decreases to around 20 nm.¹⁵ To the best of our knowledge, no systematic study has been conducted on how particle size influences self-assembly

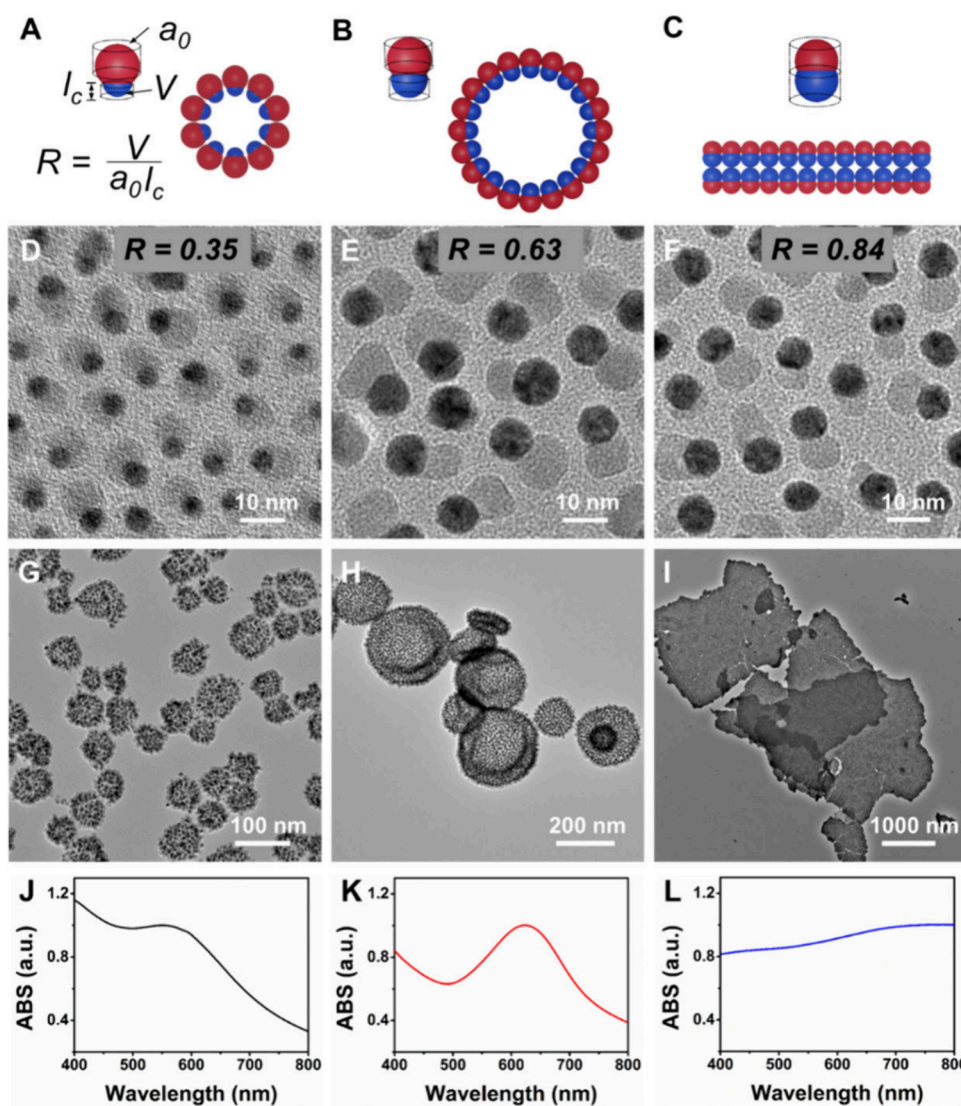


Figure 2. (A–C) Cross-sectional views of various structures from self-assembly of nanosurfactants with different Janus balance and packing parameters. TEM images of Au–Fe₃O₄ dumbbell-like nanoparticles with Janus balance of (D) 0.59, (E) 0.79, and (F) 0.92, and (G–I) their corresponding self-assemblies. (J–L) UV–vis spectra of nanosurfactants assemblies showing in (D–F).¹⁴ (Reproduced with permission from ref 14. Copyright 2020, American Chemical Society.)

behavior in bulk. However, given that many interactions occur at the submicron scale, it is speculated that size may play a significant role. This is an area worthy of future investigation.

In early studies on micron-sized JPs with amphiphilic properties, it was observed that at low concentrations, these particles assembled into micelle-like clusters, where hydrophobic faces turned inward, minimizing interaction with the surrounding environment (Figure 1a). This is similar to how surfactants with an appropriate HLB value self-assemble into micelles. As the concentration of these JPs increases, their assemblies transition into elongated, worm-like structures that closely resemble the rod-like micelles seen in surfactant systems with specific HLB values. This transition illustrates that JPs, like surfactants, can leverage Janus balance to control assembly geometry based on concentration. When concentration further increases, JPs can form unique 2D crystals as shown in Figure 1b. The particles form orientationally ordered hexagonal arrangements, with their hydrophobic hemispheres aligned facing each other. This ordered structure, resulting from directional interactions between the JPs hemispheres, is

unique because it combines translational symmetry with glassy, rotational dynamics. This structure encapsulates both the amphiphilicity and colloidal nature of JPs. Furthermore, recent studies on smaller JPs, particularly those coated with DNA, demonstrate their ability to form complex structures that resemble surfactant-based assemblies, including vesicles, bilayers, and two-dimensional layer arrangements as illustrated in Figure 1c.¹⁶ These assemblies further highlight the ability of JPs to mimic the hierarchical self-assembly behaviors typically associated with small amphiphilic molecules.

Beyond mimicking surfactants, JPs exhibit unique kinetic behaviors and structural features that set them apart. Unlike surfactant molecules, whose small size allows for rapid diffusion and rearrangement, the assembly of JPs is markedly influenced by their larger size and stronger directional interactions, which significantly alter assembly kinetics. For instance, the assembly process of JPs can be more sluggish and spatially constrained, which in turn produces assemblies in kinetically trapped states and unique geometries.¹⁰ Additionally, the unique size and shape of JPs enable the formation of

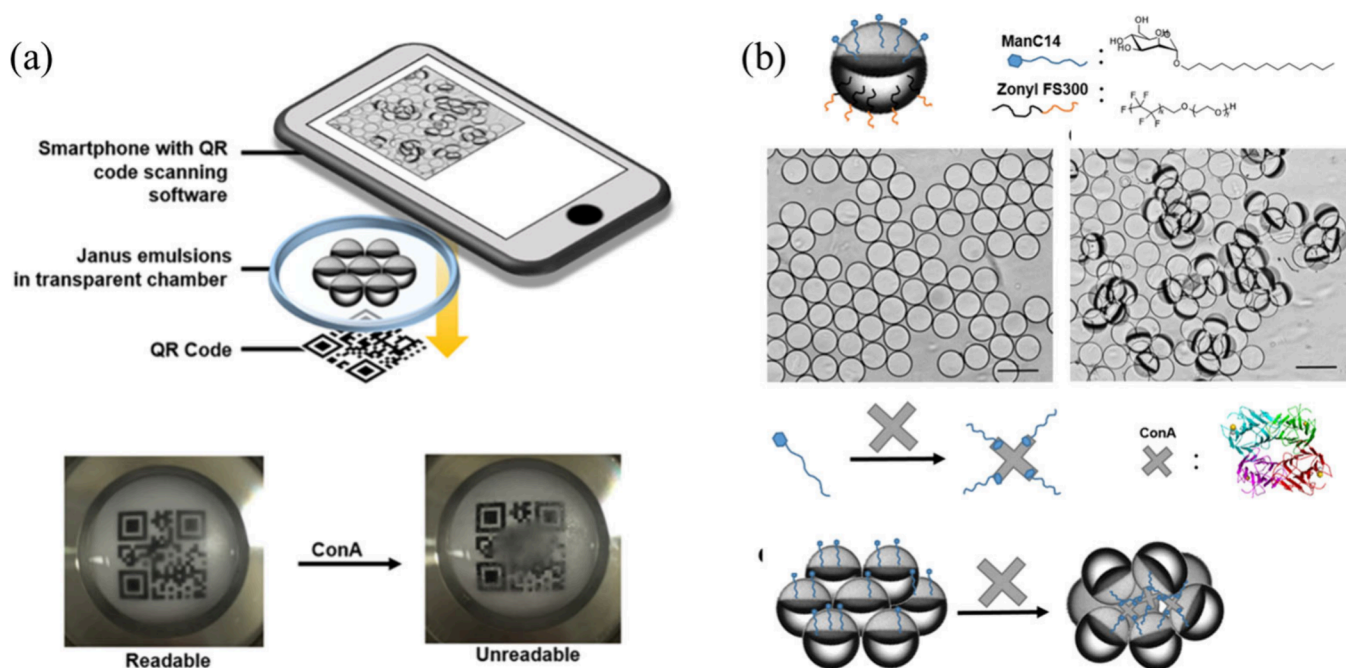


Figure 3. (a) Schematic of QR code readability for Janus emulsions in a transparent chamber (Top). Optical images showing QR code visibility before and after ConA-induced Janus emulsion agglutination, demonstrating rapid detection of analytes. Scale bar: 500 μm (Bottom). (b) Schematic and optical micrographs illustrating Janus droplets stabilized by ManC14 and Zonyl FS300. Pristine droplets exhibit a clear interface, while ConA binding induces multivalent interactions, leading to droplet agglutination and structural changes. Scale bars: 100 μm .³⁵ (Reproduced from ref 35. Available under a CC BY-NC 4.0 license. Copyright 2017, American Chemical Society.)

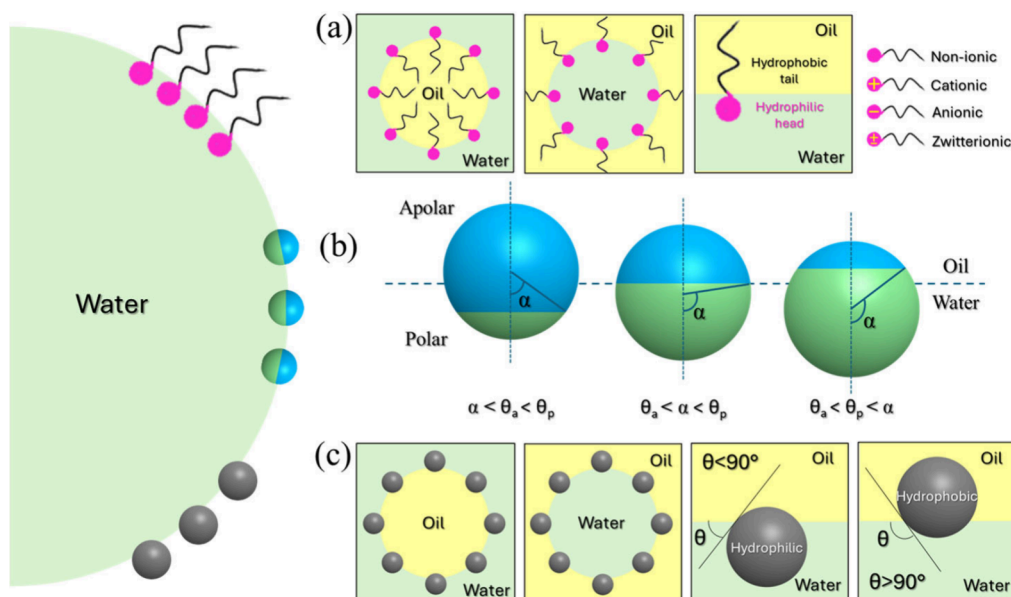


Figure 4. Emulsion stabilization by surfactants, JPs and homogeneous particles. (a) Orientation of surfactant molecules in O/W and W/O emulsions and different types of surfactants. (b) The contact angles of spherical JPs at an oil–water interface. Here, α determines the position of the boundary dividing the hydrophobic (apolar) and hydrophilic (polar) sides of the particle. The contact angles of the hydrophilic and hydrophobic sides are θ_p and θ_a , respectively. (Reproduced with permission from ref 22. Copyright 2007, AIP Publishing.) (c) Orientation of homogeneous particles in O/W and W/O emulsions and contact angle variation.

large, stable superstructures, such as vesicles and capsules, by maximizing hydrophobic lobe interactions. For example, Janus Au–Fe₃O₄ dumbbell nanoparticles (~ 20 nm) can assemble into vesicle-like and capsule structures with high stability, driven by strong hydrophobic and van der Waals forces between the Au lobes, as shown in Figure 2.^{14,15} These larger assemblies exhibit enhanced properties, including surface

plasmon resonance, where a red-shift in absorption occurs due to the close proximity of Au lobes within the assembly, a feature valuable in applications like sensing and therapeutic delivery. This capacity of JPs to form robust, reconfigurable assemblies, both surfactant-like and distinctly colloidal, highlights their versatility as tunable building blocks for advanced applications.

The assembly behavior of JPs can be harnessed for innovative sensing applications, as demonstrated in a Janus emulsion assay for the rapid and sensitive detection of *E. coli* bacteria using carbohydrate-lectin interactions (Figure 3).³⁵ In this method, Janus emulsions stabilized by a mannose-functionalized surfactant (ManC14) and a fluorocarbon surfactant (Zonyl FS300) exhibit optical clarity due to their unique morphology. Upon interaction with ConA (mannose-binding lectin, functional substitute for *E. coli*), multivalent binding takes place that induces droplet agglutination and visible tilting, creating a distinct optical contrast. This enables both qualitative detection (with the naked eye or via QR code analysis) and quantitative measurements through image processing. The rapid readout (<5 s) and self-assembled structures of Janus emulsions provide a fast, low-cost, and user-friendly approach for on-site pathogen detection, with potential applications in food safety, environmental monitoring, and portable diagnostics.

3. ADSORPTION AT INTERFACES

Surfactants can adsorb at fluid interfaces to lower interfacial tension.^{36–38} Similarly, JPs can adsorb strongly at fluid interfaces due to their unique geometry and amphiphilicity. However, the adsorption energy for JPs can be significantly higher than homogeneous particles or surfactants. Early theoretical calculations on adsorption show that when the geometry and wettabilities of the two distinct surface regions are optimized, spherical JPs can achieve adsorption energy up to three times that of homogeneous particles of the same size.³⁹ This is due to the position of JPs at the oil–water interface with one side submerged in each phase, minimizing the free energy of the system and rendering the particles effectively “trapped” at the interface.⁴⁰ For homogeneous particles, the contact angle plays a critical role in determining the particle’s affinity for the interface; intermediate contact angles around 90° yield the most stable positioning.³⁹ However, JPs, due to their distinct amphiphilic nature, may adsorb at different contact angles, enhancing their versatility in stabilizing various types of emulsions, such as oil-in-water (O/W), water-in-oil (W/O), and complex emulsions, as shown in Figure 4.^{41,42}

Similar to how the HLB value impacts surfactant adsorption at the interfaces, adsorption of JPs at the interface is governed by Janus balance, which is a critical factor in determining the particle’s equilibrium configuration at the interface.²² This balance can be manipulated by altering the ratio of the two faces or by modifying their surface chemistry, thus allowing for precise control over the particle’s behavior and adsorption characteristics. By optimizing the Janus balance, researchers can fine-tune interfacial properties to enhance stability, control droplet sizes in emulsions, and achieve selective responsiveness under external stimuli.

In addition, JPs can be synthesized in various shapes, such as rods, disks, and ellipsoids, each introducing distinct interactions at the interface.^{41–43} The geometry of these particles influences their packing density and interparticle forces, with nonspherical particles often leading to higher interfacial coverage and stability due to enhanced capillary and dipolar interactions (Figure 5a).²⁴ For instance, cylindrical JPs have shown to achieve lower interfacial tensions and more robust stability at fluid interfaces than their spherical counterparts.

The mechanical stability imparted by JPs at interfaces extends beyond mere adsorption strength to include enhanced

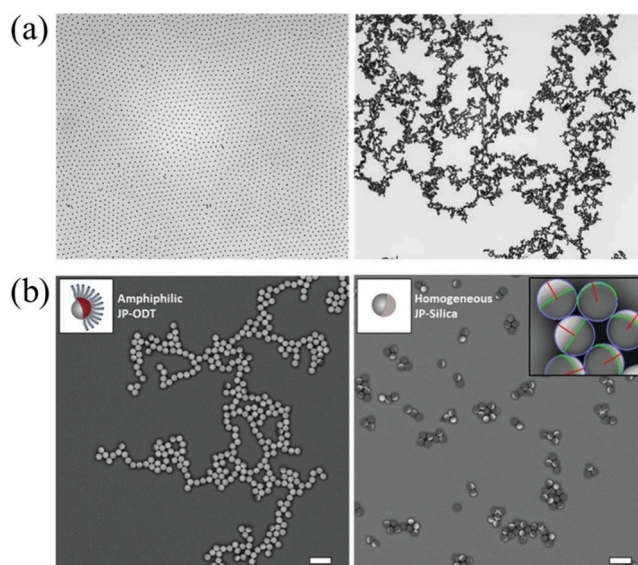


Figure 5. (a) Microstructures formed by JPs at an oil–water interface. (Left) Unmodified polystyrene particles form a hexagonal lattice due to electrostatic repulsion. (Right) Amphiphilic JPs aggregated into branched structures, driven by attractive capillary interactions.²⁴ (Reproduced with permission from ref 24. Copyright 2011, The Royal Society of Chemistry.) (b) SEM images of dried assemblies. (Left) Amphiphilic JPs (JP-ODT) form large-scale open fractal structures due to capillary attractions. (Right) Homogeneous silica particles (JP-Silica) form random, small aggregates. Insets show JP structure. Scale bars: 2 μm .²³ (Reproduced with permission from ref 23. Copyright 2018, The Royal Society of Chemistry.)

interfacial rheology. Unlike surfactants, which may migrate and reconfigure under shear forces, JPs are more resistant to displacement due to their high desorption energies and anisotropic interactions. This behavior imparts a degree of rigidity to the interface, increasing its elastic modulus and making it less susceptible to coalescence and collapse under stress.⁴⁰ Interfacial rheology studies reveal that JP-stabilized interfaces exhibit superior resistance to shear and dilational deformations compared to the surfactant-stabilized interfaces. For instance, amphiphilic silica-based Janus nanoparticles increase the interfacial shear viscosity, which enhances the stability of particle-laden emulsions against coalescence under shear. This rheological robustness is particularly beneficial in applications requiring high mechanical stability, such as enhanced oil recovery or food products, where emulsion integrity under mechanical stress is crucial.

4. EMULSION STABILIZATION

Due to the strong adsorption of JPs at interface, they can function as efficient emulsifiers. We will first consider using surfactant molecules to stabilize emulsions. As the surfactant molecules orient themselves at the interface to minimize energy, the reduction in surface tension facilitates the formation and stabilization of emulsions.⁴⁴ However, traditional surfactants do not provide high stability to emulsions because their adsorption is relatively weak and reversible.⁴⁵ This means that while surfactants can temporarily prevent coalescence, their tendency to desorb from the interface over time leads to gradual breakdown and destabilization of emulsions. The reversible nature of surfactant adsorption also implies that achieving stable emulsions often requires relatively

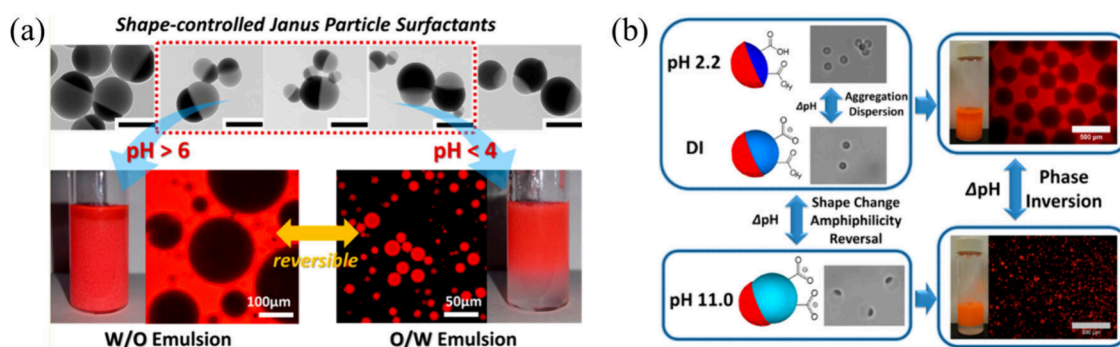


Figure 6. pH-responsive JPs. (a) Reversible switching of emulsion types between W/O at $pH > 6$ and O/W at $pH < 4$ by adjusting the pH. Self-stratifying JPs create a hydrophobic monolayer atop binder particles, that improves water contact angle and surface resilience in coatings.⁴⁹ (Reproduced with permission from ref 49. Copyright 2017, American Chemical Society.) (b) pH-responsive JPs exhibiting shape and amphiphilicity changes, enabling reversible aggregation and dispersion at different pH levels and inducing phase inversion in emulsions. Insets show confocal microscopy images of emulsion morphologies. Scale bar: 500 μm .⁵⁰ (Reproduced with permission from ref 50. Copyright 2014, American Chemical Society.)

high concentrations, as large amount surfactants are needed at the interface to maintain droplet stability.

In contrast, JPs exhibit much stronger desorption energies at fluid interfaces, thanks to their size and structural asymmetry.⁴⁶ By anchoring themselves at the interface with each side in its preferred phase, JPs effectively "lock" into the place, creating a robust barrier that prevents droplet coalescence and stabilizes the emulsion.²² Unlike surfactants, which stabilize emulsions through a dynamic equilibrium of adsorption and desorption, JPs stabilize emulsions through their strong, almost irreversible adsorption. Therefore, JPs can stabilize emulsions more effectively than molecular surfactants.³⁹ In addition, the high adsorption energy also makes JPs capable of stabilizing emulsions more efficiently at lower concentrations than surfactants. Since JPs do not need to be continuously replenished at the interface, lower concentrations are sufficient to maintain stability, whereas surfactants typically require higher concentrations to sustain the dynamic balance needed for droplet stabilization. This efficiency makes JPs more cost-effective and desirable for applications requiring minimal additive usage, such as pharmaceuticals and environmentally sensitive products.

Another fundamental distinction is that JPs, due to their amphiphilic structure, are well-suited to stabilize both simple and complex multiphasic systems.^{30,47,48} Surfactants, on the other hand, are often limited in their performance in complex emulsions because of their relatively low adsorption energy and sensitivity to environmental changes. The robust attachment of JPs to the interface allows them to stabilize emulsions under conditions that would be challenging to traditional surfactants, such as in the presence of salts, variations in pH, or high temperatures.

Furthermore, JPs can be designed to respond to various environmental stimuli, significantly influencing the assembly or emulsion behavior. For example, pH-responsive JPs can even alter their amphiphilicity and cause emulsion inversion, a phenomenon where an oil-in-water emulsion switches to a water-in-oil emulsion (or vice versa) in response to a change in pH (Figure 6).^{49,50} The JPs are synthesized through seeded emulsion polymerization followed by acid hydrolysis, resulting in a biphasic structure with one side rich in styrene (hydrophobic) and the other in acrylic acid (pH-sensitive and hydrophilic). At low pH, the acrylic acid side remains protonated and less swollen, resulting in an overall hydro-

phobic character and leading to W/O emulsion stabilization. At high pH, ionization of the acrylic acid groups induces significant swelling, transforming the particle shape and increasing the hydrophilicity, which facilitates oil-in-water O/W emulsion stabilization. Another example is that Au-Fe₃O₄ JPs enable triggered release of encapsulated payloads via IR-laser heating, solvent changes, and pH adjustments. The strong interfacial activity and dual plasmonic-magnetic properties of JPs allow precise, controlled release, making them ideal for drug delivery without the need for extra surfactants.^{14,15} The ability to adapt to different environments provides JPs with a clear advantage over surfactants, especially in the systems that require controlled changes.

Nonspherical forms, such as Janus dumbbells, also serve as effective emulsifiers.⁴⁶ Here, the ratio of the lobe sizes impacts surface activity, potentially stabilizing nonspherical emulsion droplets by jamming the dumbbells at the interface, a mechanism that acts on time scales faster than droplet relaxation. Amphiphilic Janus nanosheets, particles with plate-like structures, have also been shown to be effective emulsifiers.⁵¹ Their flat geometry and amphiphilic nature enable them to reduce droplet size effectively. In emulsion polymerization, increasing the concentration of these nanosheets reduces droplet size further, resulting in well-dispersed and monodisperse particles, as demonstrated with polystyrene particle emulsions. JPs can also adsorb strongly at polymer-polymer interfaces and the resulting adsorption remains stable without forming micelles, even under high temperature and shear conditions.⁹ In mini-emulsion polymerization, JPs eliminate the need for additional additives.⁵² This efficiency allows for controlled, monodisperse particle formation, making JPs highly effective stabilizers in emulsion polymerization.

5. DISPERSANTS FOR EMULSION POLYMERIZATION

JPs, with their amphiphilic and asymmetric structure, present distinct advantages as dispersants in emulsion polymerization. Traditionally, block copolymers are employed for their ability to self-assemble and stabilize emulsion systems.^{53–56} This behavior arises from their amphiphilic composition, where hydrophobic and hydrophilic blocks segregate to form stable micellar structures.⁵⁶ JPs extend this functionality by offering a higher degree of control over surface activity and mimicking the self-assembly tendencies of block copolymers but with enhanced stability and versatility.^{23,57,58} While both JPs and

Table 1. Comparison of Amphiphilic JPs with Surfactants in Terms of Assembly Structure, Adsorption, Emulsion Stabilization, Functionality, and Limitations

Feature	Amphiphilic JPs	Surfactants	References
Behavior in Bulk	Forms complex structures (micelles, worms, crystalline assemblies)	Forms micelles or bilayers depending on concentration	1–3, 8, 9
Adsorption Energy	>1000 kT, preventing desorption	~kT, reversible adsorption	38, 39
Interface Stability	More stable emulsions	Prone to coalescence	22, 39, 45, 46
Concentration Requirement	Low concentration is required	High concentrations needed	39, 52, 59
Functionality	Multifunctionality	No additional functionality	14–18, 35, 46–49
Cost	Not commercially available	Inexpensive and widely available	45, 66

diblock copolymers can stabilize emulsions, JPs have higher surface activity at interfaces, resulting in higher adsorption energy than diblock copolymers.

Walther et al. discovered a unique template-assisted method to fabricate JPs using microphase-separated triblock terpolymers composed of polystyrene (PS), polybutadiene (PB), and poly(methyl methacrylate) (PMMA).⁵² By selectively cross-linking PB domains and hydrolyzing PMMA, they obtained amphiphilic JPs with hydrophobic PS on one side and hydrophilic poly(methacrylic acid) (PMAA) on the other, measuring ~100 nm in diameter. These JPs demonstrated effectiveness as dispersants for emulsion polymerization, offering important advantages over diblock copolymers.

Additionally, JPs allow precise control over particle size and polydispersity. While achieving low polydispersity and controlled particle sizes with diblock copolymers often requires high concentrations, JPs achieve similar or better results at lower concentrations, making them more efficient and cost-effective dispersant for emulsion polymerization.^{52,59} Overall, JPs represent more advantages over traditional diblock copolymers, particularly in terms of surface activity and stabilization efficiency. These properties make JPs an attractive option for industrial emulsion polymerization, offering both technical improvements and economic benefits. The major differences between JPs and surfactant molecules are summarized in Table 1.

6. JPS AS COATING MATERIALS

Coating materials are integral to daily life, from household paints to traffic markings. Beyond aesthetics, coatings provide essential surface protection. Over the past 60 years, the coating industry has transformed significantly, driven by technological advances, regulatory changes, and cost pressures. In many regions, regulations now limit volatile organic compounds (VOCs) in coatings, and environmentally conscious consumers increasingly seek eco-friendly products. However, many coating systems developed in academic laboratories face challenges in industrial scalability due to high material and processing costs. Developing a successful coating product requires a deep understanding of both current technologies and market demands. In this context, we will examine the cases where JPs may enhance coating performance by adding functionality, reducing raw material usage, and improving sustainability. We will also briefly discuss the current challenges facing the coating industry and how novel materials like JPs may help address these issues.

Coating materials have long relied on surfactants and dispersants, which play critical roles in stabilizing and enhancing the properties of formulations. Dispersants, a specialized type of surfactant, are specifically designed to stabilize pigment and filler particles within coatings.⁶⁰ Given

their similarity to these molecular counterparts, JPs may significantly enhance coating materials. At the same time, coating materials are remarkably diverse, encompassing a wide range of chemistries, such as acrylic, polyurethane, and epoxy systems, and formats, including waterborne, solvent-borne, and solvent-less powder coatings. Additionally, these coatings serve a variety of applications, from wood stains to architectural wall paints and industrial metal coatings. Recognizing these differences is essential when discussing the fundamental composition and applications of coatings.

Recent developments in JP-based coatings have shown promising outcomes across multiple areas, including low-VOC formulations, as well as antifouling and anti-icing technologies. Here, we highlight the latest advancements in applying JP research to coating materials. JPs offer several advantages over conventional colloidal particles due to their dual chemical properties. For example, amphiphilic JPs, which can be considered as colloidal surfactants, exhibit unique self-stratification behavior, allowing them to spontaneously organize at interfaces when mixed with homogeneous binder particles (Figure 7a).^{25,41} This property enables JPs to control surface characteristics, which is especially valuable for enhancing water resistance in waterborne coatings.

We will explore several cases where JPs have improved coating performance, focusing on low-VOC, anti-icing, and antifouling coatings. For each type, we will provide background information, briefly discuss JP synthesis, and examine the applications of these particles in enhancing coating functionality.

6.1. Low-VOC Coatings. The coating industry has undergone considerable changes over recent decades due to stricter environmental regulations and concerns about pollution. Most notably, the Clean Air Act (CAA) aims to limit hazardous air pollutants (HAPs) such as VOCs, which contribute to air pollution and pose health risks such as respiratory diseases and potentially cancer.^{61–63} VOCs, the compounds that readily vaporize at room temperature, are commonly found in traditional oil-based paints and solvent-based coatings. To address these challenges, the industry has increasingly developed waterborne coatings that use water as the main solvent, greatly reducing the amount of VOCs present in the system.⁶⁴ However, these often suffer from performance drawbacks, including reduced water resistance, adhesion, suspension stability, and film formation. Conventional solutions, like multilayered primer and topcoat systems, are time-consuming and not cost-effective.⁶⁵

Acrylic-based emulsion polymerization is central to waterborne coatings that significantly reduced VOC emissions, from 700 g/L in the 1940s to around 50 g/L by the 2010s.⁶⁶ Despite these environmental advances, waterborne coatings continue to face issues related to water resistance, surface hardness, and overall stability due to their intrinsic formulation in water.

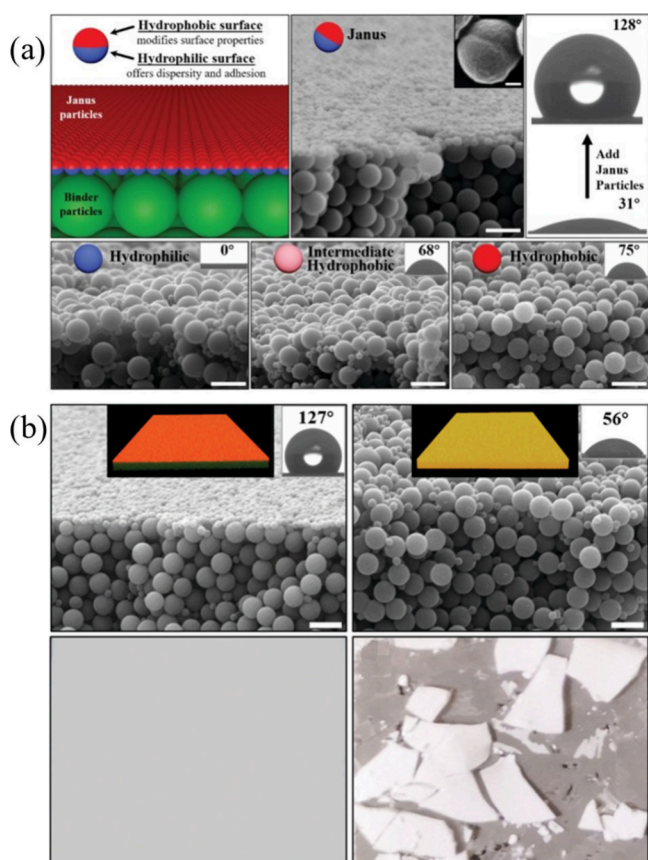


Figure 7. (a) Self-stratification of amphiphilic JPs forms a hydrophobic surface layer, enhancing water resistance and durability, while homogeneous particles fail to stratify. Insets show confocal images and contact angles of the coatings. (b) SEM images of coating films after tetrahydrofuran rinsing: JP films show self-stratification, while hydrophobic homogeneous particle films do not. Insets: confocal fluorescent images and post-rinse contact angles. Scale bars: 2 μm . (Top). Photos of coating films with JPs and hydrophobic homogeneous particles after solvent rinsing (Bottom).²⁵ (Reproduced with permission from ref 25. Copyright 2020, Wiley-VCH.)

Recent developments in amphiphilic JPs offer a promising approach for addressing these challenges in waterborne coatings.

Li et al. synthesized amphiphilic JPs (~ 500 nm) and directly incorporated them as an additive into a waterborne coating system.²⁵ These particles rapidly self-stratified at the surface to form a densely packed monolayer with minimal quantities required. The hydrophobic lobes faced the air, enhancing water resistance, while the hydrophilic lobes oriented toward the binder particles, promoting adhesion. This self-stratification mechanism improved the water contact angle and durability of the coating, creating a water-resistant layer that withstands solvent rinsing (Figure 7b). Waterborne coatings with self-stratified JPs exhibited enhanced durability, surface hardness, and reduced tackiness compared to conventional systems. This innovative approach enables selective tuning of JP chemistry for optimized surface properties, while binder chemistry is optimized for bulk properties, addressing common waterborne coating limitations without major formulation changes.

In another study, He et al. utilized amphiphilic JPs to improve the mechanical properties of waterborne bonded solid lubricating coatings (BSLCs).⁶⁷ Dumbbell-shaped JPs, with one lobe composed of rigid silica and the other of soft polytetrafluoroethylene (PTFE), were synthesized. The hydrophilic silica allowed for homogeneous JP dispersion in a waterborne epoxy BSLC system that also contained hydrophobic PTFE as a lubrication additive. By incorporating JPs, lubrication and wear resistance improved due to PTFE's flexibility and silica's rigidity, reducing surface wear. Additionally, PTFE's hydrophobicity and low surface energy enhanced the coating's hydrophobic and oleophobic properties, supporting BSLC's eco-friendly applications by reducing VOC reliance.

The integration of JPs into low-VOC coatings provides a promising pathway toward durable, eco-friendly solutions, addressing both environmental and performance needs. Through self-stratification and improved material properties, JPs allow waterborne coatings to match or exceed the

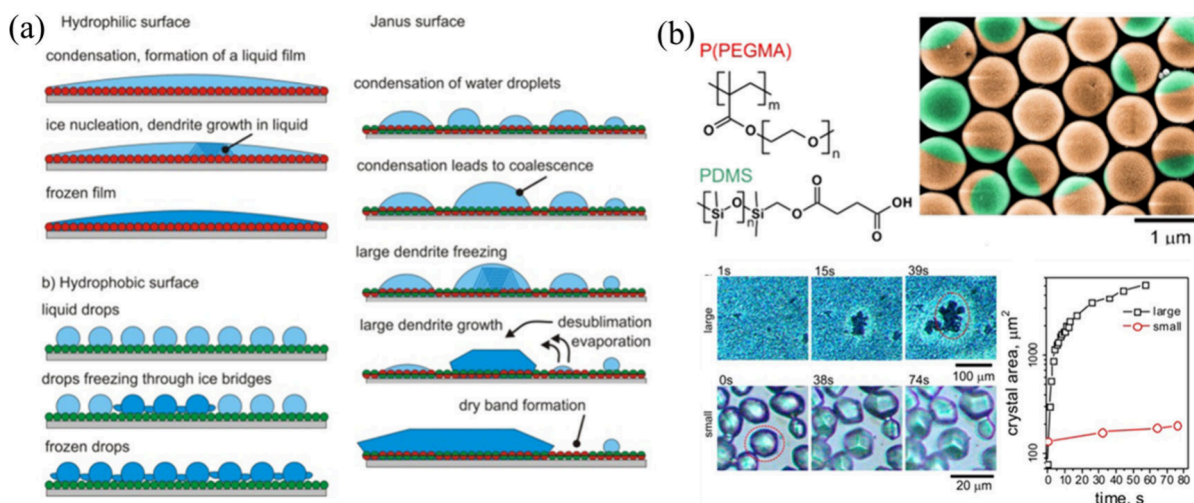


Figure 8. (a) Anti-icing mechanisms on hydrophilic, hydrophobic, and Janus surfaces, with Janus surfaces reducing ice adhesion through controlled droplet coalescence and dry band formation. (b) P(PEGMA)/PDMS JPs exhibiting distinct hydrophilic and hydrophobic domains enable controlled water condensation and ice nucleation. Time-lapse imaging shows the rapid growth of large ice dendrites on Janus surfaces, significantly enhancing anti-icing performance through efficient water droplet management and reduced ice adhesion.⁶⁸ (Reproduced with permission from ref 68. Copyright 2016, American Chemical Society.)

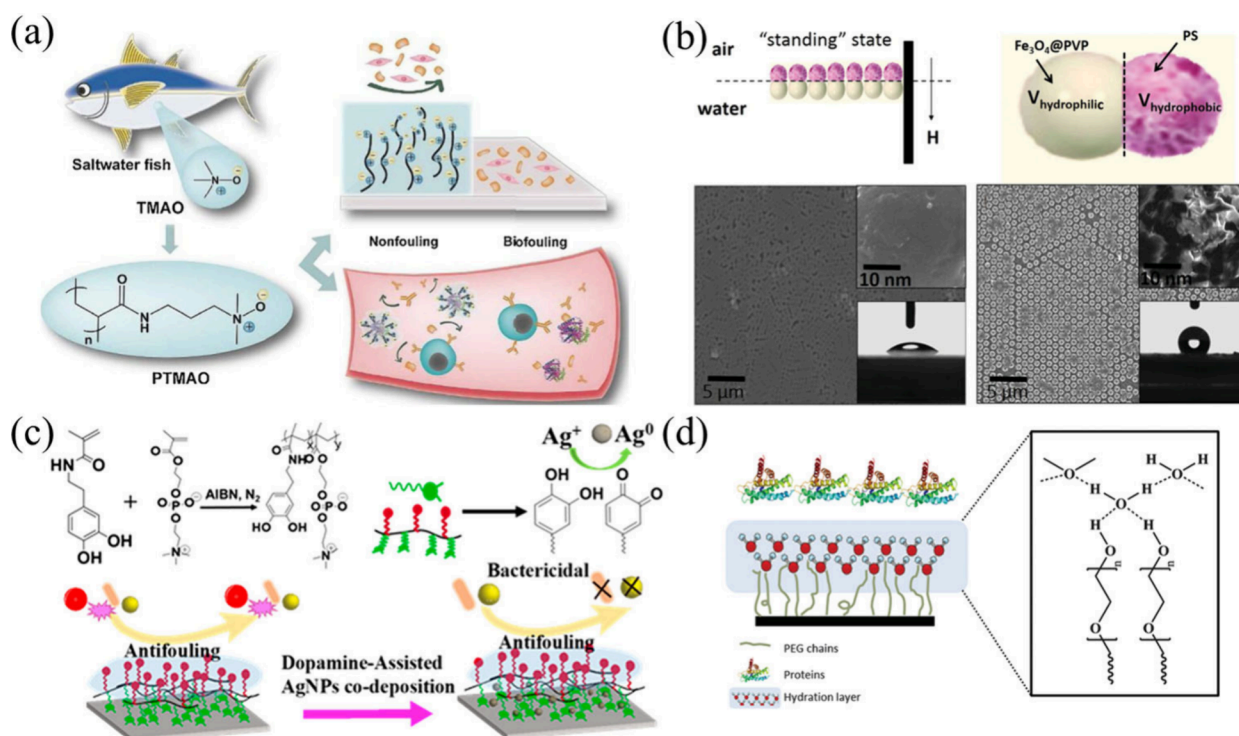


Figure 9. (a) Zwitterionic polymer PTMAO, inspired by saltwater fish, forms a stable hydration layer that prevents biofouling.⁷⁵ (Reproduced with permission from ref 75. Copyright 2020, Elsevier.) (b) Magnetically responsive Fe_3O_4 @PVP/PS JPs, exhibiting distinct hydrophilic and hydrophobic regions, assemble in a "standing" orientation at the air–water interface, facilitating controlled orientation and self-assembly.⁷⁸ (Reproduced with permission from ref 78. Copyright 2016, Wiley-VCH.) (c) Dopamine-assisted codeposition of zwitterionic copolymers and AgNPs creates a dual-function antimicrobial and antifouling surface, enhancing biofilm resistance.⁷³ (Reproduced with permission from ref 73. Copyright 2024, American Chemical Society.) (d) Schematic of antifouling surfaces using PEG chains to form a hydration layer that resists protein adsorption through steric repulsion and hydrogen bonding.⁷¹ (Reproduced from ref 71 with permission from the Royal Society of Chemistry.)

performance of traditional solvent-based systems, meeting stringent regulatory requirements and rising demand for sustainable products.

6.2. Anti-icing Coatings. JPs show immense potential in developing anti-icing coatings due to their unique ability to control water condensation and ice formation. With distinct hydrophilic and hydrophobic regions, JPs, when applied to a coating surface, create heterogeneous surfaces that reduce ice adhesion and delay ice nucleation (Figure 8a). This dual functionality is particularly valuable in passive anti-icing coatings, providing energy-efficient alternatives to conventional deicing methods.

Kirillova et al. demonstrated that hybrid JPs with hydrophilic and hydrophobic sides form heterogeneous surfaces, effectively minimizing ice adhesion (Figure 8b).⁶⁸ The synthesis is based on a wax-based method. These JPs cause water to condense on the hydrophilic regions, pinning it at the boundary with the hydrophobic areas as temperatures drop. This creates irregular clusters that freeze into large, unstable ice dendrites, which adhere weakly to the surface, reducing ice adhesion to as low as 56 kPa. This approach offers a highly effective, passive solution for anti-icing coatings, significantly reducing the need for mechanical or chemical deicing.

Beyond ice adhesion reduction, JPs significantly contribute to preventing or delaying ice nucleation. Schwarzer et al. showed that hybrid JPs with dual-modality surfaces effectively delayed ice nucleation when supercooled water droplets impacted the coated surface.⁶⁹ Experimental surfaces with these coatings exhibited prolonged delays in ice nucleation, even after 100 impacts of supercooled droplets. This effect is

attributed to rapid dewetting and contraction of droplets on impact, minimizing the contact area and reducing nucleation sites—a benefit particularly useful for applications where ice buildup disrupts operations.

Further advancing JPs' role in anti-icing, Kirillova et al. synthesized platelet JPs with hydrophilic and hydrophobic polymer shells using a scalable approach based on surface-initiated Atom Transfer Radical Polymerization (ATRP).⁷⁰ These amphiphilic particles enhance ice-prevention properties: the hydrophilic side interacts with water molecules, while the hydrophobic side repels them, reducing ice formation. Additionally, these JPs stabilize emulsions and polymer blends, positioning them as candidates for multifunctional coatings that combine anti-icing with self-healing or corrosion resistance.

6.3. Antifouling Coatings. Antifouling coatings are essential in marine and biomedical applications to prevent unwanted biological material accumulation on surfaces.^{71,72} Zwitterionic polymers have been extensively studied as effective antifouling agents (Figure 9a).^{73–75} Recently, JPs have emerged as a promising alternative for antifouling coatings due to their unique bifacial structure and versatile properties.^{76,77} The combination of hydrophilic and hydrophobic groups allows JPs to create certain patches with high contrast of hydrophobicity at interfaces, disrupting the adhesion of fouling organisms. This feature makes JPs ideal candidates for applications where surface contamination by biological materials, such as proteins, bacteria, or marine organisms, is a concern.

Wang et al. expanded on JP applications by introducing a novel method for synthesizing amphiphilic JPs through a combination of diffusion-induced phase separation and magnetically driven dewetting.⁷⁸ The amphiphilicity, shape anisotropy, and magnetic properties of these JPs enable their synergistic self-assembly into large-area, amphiphilic monolayers. These structures exhibit controlled orientation and high-order regularity, particularly under applied magnetic fields (Figure 9b). The assembled monolayers display dual wettability and have potential applications in antibioadhesion materials and antifouling surfaces, as hydrophilic surfaces promote cell adhesion, while superhydrophobic surfaces inhibit cell adhesion. This paradigm offers a new direction in JP design for antifouling coatings and advanced surface applications, although further research is needed to understand the role of factors like shape anisotropy, amphiphilicity, and magnetic fields.

Despite the promise of JPs for antifouling applications, challenges remain. A primary obstacle is scaling up JP production while preserving their bifacial structure. Additionally, ensuring long-term JP stability in various environmental conditions is crucial for maintaining their effectiveness in real-world applications. Environmental factors such as temperature and chemical exposure can potentially impact JP-based coatings' performance and durability.^{16,79} Nonetheless, JPs represent an important advancement in antifouling technology, providing innovative solutions for industries from marine transportation to biomedical devices.

Both zwitterionic polymers and JPs have been the focus of studies aimed at enhancing antifouling coatings. For instance, coatings combining zwitterionic polymer brushes with silver nanoparticles have been shown to reduce the adhesion of both Gram-positive and Gram-negative bacteria while preventing protein adsorption, such as bovine serum albumin (Figure 9c).⁷³ JPs have further advanced antifouling innovations by creating chemically heterogeneous surfaces that lower bacterial adhesion and improve coating durability under real-world conditions. Studies show that JPs grafted with antifouling polymers like poly(ethylene glycol) (PEG) or polydimethylsiloxane (PDMS) can significantly reduce bacterial settlement and biofilm formation, making them a promising option for scalable antifouling coatings (Figure 9d).^{71,79} Through such advancements, JPs offer a new pathway for antifouling technologies, helping address long-standing challenges in diverse applications.

7. JPS VS ZWITTERIONIC POLYMERS

Building upon the versatile applications of JPs in advanced functional coatings, it is quite essential to explore how these materials can be compared to other high-performance alternatives. Zwitterionic polymers, renowned for their exceptional antifouling capabilities and biocompatibility, represent a compelling parallel to JPs in terms of amphiphilicity and environmental responsiveness. Both materials have demonstrated remarkable potential in creating functional and sustainable coatings.^{73,80–84} This section highlights the similarities and distinctions between the two, with a special focus on their respective strengths and application domains.

Zwitterionic polymers are a subset of ionic polymers that exhibit both cationic and anionic groups along their polymer backbone, yet maintain an overall neutral charge or a charge balance, similar to JPs that feature distinct regions with contrasting surface properties, often one hydrophobic and the

other hydrophilic.^{85–87} This characteristic sets them apart from traditional polyelectrolytes, which are either cationic or anionic.^{88,89} While zwitterionic polymers and JPs may seem structurally distinct, they share several similarities that make them valuable in similar applications.^{90,91} Both materials exhibit amphiphilic properties and can self-assemble into complex structures under certain conditions.^{92,93} Their versatility in responding to environmental stimuli, such as pH, ionic strength, or temperature, makes them attractive for use in coatings, biomedical devices, and antifouling surfaces.^{94,95} Zwitterionic polymers are particularly renowned for their antifouling capabilities. Their charge balance and hydration layers prevent the adsorption of proteins and cells on surfaces, making them ideal for medical devices and marine applications.^{96–98}

JPs, although less extensively studied for antifouling, offer unique advantages by combining different functionalities on a single particle. For instance, amphiphilic JPs can reduce bacterial adhesion due to their contrasting hydrophobic and hydrophilic regions, while their chemical heterogeneity provides tunable antifouling properties. Studies have demonstrated that JPs can lower bacterial adhesion under static conditions when compared to fully hydrophobic or native surfaces, indicating their potential as robust antifouling materials.

Both zwitterionic polymers and JPs hold significant promise for biomedical applications. Zwitterionic polymers are being extensively explored in coatings for medical devices to prevent protein adsorption and resist cell adhesion. Their ability to form stable hydration layers also imparts anti-inflammatory and antibacterial properties. Similarly, JPs, with their multifunctional surfaces, can be tailored for targeted drug delivery systems or to stabilize emulsions in medical formulations. Their unique ability to interact with both hydrophilic and hydrophobic environments enables self-assembly into complex nanostructures, enhancing drug encapsulation and facilitating stimuli-responsive controlled delivery. Notably, limited research has explored the integration of zwitterionic polymers with JPs. Future studies could focus on developing JPs with zwitterionic functionalities, potentially unlocking even greater performance in biomedical applications.

8. FUTURE DIRECTIONS

This review reinforces the comparative advantages of JPs over traditional surfactants and examines their expanding role across various applications, from emulsion stabilization to advanced coating solutions. Their colloidal nature, combined with their anisotropic structure, enables unique assembly behaviors and desorption energies that set them apart, particularly in terms of stability at interfaces. This high-energy desorption, surpassing that of surfactants, fosters long-term stability in emulsions, requiring lower JP concentrations than surfactants for effective stabilization. Additionally, JPs' ability to assemble in diverse structural configurations in bulk media and their responsiveness to external stimuli like pH shifts underscore their versatility.

The potential of JPs in coating materials, particularly low-VOC, anti-icing, and antifouling formulations, highlights their role as environmentally friendly alternatives aligned with regulatory demands for sustainable products. Self-stratification, a unique property of JPs when mixed with homogeneous particles, enables them to assemble at interfaces, enhancing the durability and water resistance of coatings. By integrating JPs

into waterborne coatings, the industry can mitigate common issues such as low adhesion and poor water resistance without extensive formulation modifications, making these materials a cost-effective option for large-scale applications.

However, the challenges of scaling up JP production while retaining consistent morphology and properties remain significant. Achieving stable, high-quality emulsions or coatings across varying environmental conditions necessitates further research into optimizing synthesis methods. Additionally, a more in-depth understanding of JP interactions within industrial systems is essential for maximizing their commercial potential.

The multifunctional capabilities of JPs position them as valuable assets in areas requiring adaptive materials. For instance, incorporating JPs in anti-icing and antifouling applications can reduce environmental impact and maintenance costs in marine and biomedical settings. Additionally, JPs with specifically crafted hydrophobic and hydrophilic domains offer promising pathways for next-generation coating materials capable of dynamic performance in challenging conditions. Future research may adopt a holistic approach, merging computational modeling with experimental insights to accelerate JP development, bridging the gap between molecular surfactants and colloidal particles. As JPs move from lab-scale investigations to commercial applications, interdisciplinary collaboration will be crucial, combining expertise in particle synthesis with application-focused engineering to explore new possibilities.

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Author Contributions

All authors contributed to the writing and revisions of the manuscript. Utsav Kumar Dey: Contributed to the writing of all sections of the manuscript. Serkan Demirci: Contributed to the writing of antifouling coatings. Ricardo Ortega: Contributed to writing the writing of coating materials. Thamer Rawah: Contributed to the writing of anti-icing coating. Aneeba Chaudary: Contributed to the writing of anti-fouling coatings. Fei Liu: Contributed to the writing of assembly at bulk. Zhengtao Yang: Contributed to writing the writing of coating materials. Bingrui Huang: Contributed to writing the writing of Low-VOC coating. Shan Jiang: Structured the manuscript, wrote the introduction, and contributed to writing and revising all sections of the manuscript.

Notes

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